miniRUEDI Symposium

18–20 November 2025

Eawag, Überlandstrasse 133, 8600 Dübendorf, Switzerland

Tuesday, 18 Nov 2025

9:00 – 9:30	Coffee and snacks
9:30 – 9:40	Welcome
9:40 – 10:00	Introduction
10:00 – 10:40	Presentations
10:40 – 11:10	Break
11:10 – 12:10	Presentations
12:10 – 14:00	Lunch at Eawag
14:00 – 15:00	Presentations
15:00 – 15:30	Break
15:30 – 16:30	Presentations

Wednesday, 19 Nov 2025

9:00 – 9:30	Coffee and snacks
9:30 – 10:30	Presentations
10:30 – 11:00	Break
11:00 – 12:00	Presentations
12:00 – 14:00	Lunch at Eawag
14:00 – 15:00	Presentations
15:00 – 15:30	Break
15:30 – 16:30	Presentations
18:00	Group dinner (open end)

Thursday, 20 Nov 2025

9:00 – 9:30	Coffee and snacks
9:30 – 10:30	Presentations
10:30 – 11:00	Break
11:00 – 11:40	Presentations
12:00 – 14:00	Lunch at Eawag
14:00 – 15:00	miniRUEDI hands-on, lab tour

XY Nov 2025, HH:MM–HH:MM

Seasonality of Excess Air Signatures Under Different Flow Conditions

Christoph West, Nicholas Thiros, Werner Aeschbach, Payton Gardner

Institute of Environmental Physics Heidelberg

'- hydrograph of a high altitude fractured shale bedrock mountain aquifer in the East River watershed, Colorado, US driven by seasonal snowmelt patterns - results in two flow periods: recharge (peak snowmelt, water table rise), baseflow (no snowmelt/recharge events, water table lower at a base level) - bulk noble gas measurements during recharge period show signature of excess air, while measurements during the baseflow period show a degassed signature - subsurface production of CO_2, N_2, and CH_4 have been observed previously resulting from bedrock weathering of the shale - Possible explanation: - Higher water table (during the recharge period) corresponds to higher hydrostatic pressure, which can dilute more gas into the groundwater, thus showing excess air in the noble gases - Lower water table (during baseflow) corresponds to lower hydrostatic pressure, thus oversaturated gas exsolves continuously and leads to the degassed signature

XY Nov 2025, HH:MM–HH:MM

Impact river work entangled with gas data

Théo Blanc, Friederike Currle, Morgan Peel ...etc... Brunner Roki Schilling

Universiy of Neuchatel / Eawag

Entangle the travel time and mixing ratios from freshly infiltrating river water into groundwater. Use environmental gas tracers miniRuedi and Rad 7. Binary mixing models coupled with excess air model. The travel times change drastically because of riverwork and high discharge event.

XY Nov 2025, HH:MM–HH:MM

Photosynthesis and productivity rates in the Eastern Tropical Pacific Ocean over the 2023-2025 El Niño cycle

Hedy M. Aardema, Hans A. Slagter, Lena Heins, Ralf Schiebel, Gerald H. Haug

AFFILIATIONS

DRAFT The El Niño-Southern Oscillation (ENSO) is a natural climate variability with strong fluctuations in the ocean environment that impact biota and productivity. This provides a natural laboratory for testing the response of productivity to rapid ocean warming. We conducted 4 expeditions in the Eastern Tropical Pacific to study the impact of this climatic variability on phytoplankton photophysiology and productivity. On the S/Y Eugen Seibold the underway set-up includes a LabSTAF active fluorometer (Chelsea, UK) and miniRUEDI mass spectrometer (Gasometrix, Switzerland) to measure photophysiological rates and net community productivity rates, respectively.

XY Nov 2025, HH:MM–HH:MM

Tracking CO_2 injection to an aquifer for CCS in Iceland

Matthias S. Brennwald, Jonas Simon Junker, Chuan Wang, Rolf Kipfer, Anne Obermann, Martin Voigt, and Alba Zappone

Eawag

Mineral sequestration of CO_2 in basalt aquifers is a promising pathway for permanent carbon storage. Within the DemoUpStorage R&D project, CO_2 was dissolved in saline groundwater and co-injected with helium as an inert tracer into a deep basalt aquifer at a pilot injection site near Helguvik, Iceland. We tested a combination of two novel geochemical and geophysical tools to monitor the subsurface CO_2 dynamics. Real-time, onsite monitoring of dissolved gases was conducted over one year using a gas-equilibrium membrane-inlet mass spectrometer (GE-MIMS). Electrical resistivity tomography (ERT) was used to track changes in aquifer resistivity before and after injection. GE-MIMS measurements revealed a clear increase in He concentrations downstream of the injection point, confirming fluid transport. However, no corresponding CO_2 increase was observed, indicating substantial CO_2 retardation relative to the fluid transport. No anomalies in He or CO_2 were detected in the overlying freshwater aquifer, indicating minimal upward migration. ERT data showed localized resistivity decreases suggesting basalt dissolution near the injection borehole. Together, GE-MIMS and ERT provided complementary, cost-effective insights into CO_2 transport, reaction, and storage processes in the basalt aquifer. These tools enhance monitoring and assessment capabilities, supporting the development of safe and effective geological CO_2 storage.

XY Nov 2025, HH:MM–HH:MM

Origin of fluid emissions along crustal to lithospheric faults in an extensional setting, Betic Cordillera, Spain

Bérénice Cateland, Anne Battani, Matthais Brennwald, Benjamin Lefeuvre, Nicolas E. Beaudoin, Frédéric Mouthereau, Rolando Burga, Jose Benavente, Magali Pujol

Université de Pau et des pays de l'Adour

The Betic Cordillera (SE Spain) has experienced a complex geodynamic history involving crustal thinning related to slab retreat. This deep lithospheric dynamic was accompanied by alkaline to calc-alkaline volcanism and the exhumation of metamorphic domes. These processes feed an active fluid system, the respective contributions of which remain poorly constrained. However, understanding and quantifying these deep fluid circulations has major implications for geothermal exploration and natural gas migration. Several crustal to lithospheric-scale faults in the Internal Betics have accommodated most of the deformation. These active structures (Lorca earthquake, Mw 5.2) are associated with significant hydrothermalism, as indicated by water temperatures reaching up to 65 °C, and may act as preferential pathways for fluid and heat migration. We sampled fluids (water and gas when bubbles were present) from 14 thermal springs with temperatures ranging from 15 to 65 °C. Gas species were measured in the field using a portable mass spectrometer (MiniRUEDI) along these strike-slip faults. The Cádiz-Alicante Fault releases fluids rich in N2 (86–96%), with CO2 contents between 1 and 5%, and O2 between 0.5 and 10%. Along the Alpujarras Fault, CO2 dominates (63–91%) with lower amounts of N2 (5–28%). The lithospheric Carboneras Fault emits fluids rich in N2 (91%), with low proportions of O2 (3%) and CO2 (3%). Alhama de Murcia, the most active fault of the system, mainly releases CO2 (52%) and N2 (43%). This result helps us in discussing fluid origin in this extensive context.

XY Nov 2025, HH:MM–HH:MM

Year-Long miniRUEDI Gas Time Series Reveal Shutdown-Driven Excess Air and Water-Quality Impacts

Jared van Rooyen, Yama Tomonaga, Matthias S. Brennwald, Rolf Kipfer, Oliver S. Schilling

Eawag

Managed aquifer recharge (MAR) operations can unintentionally modulate subsurface gas dynamics, with consequences for redox conditions and groundwater quality. In this study, we deploy a portable mass spectrometer (miniRUEDI) in an observation well at the Hardwald MAR scheme near Basel, Switzerland, and acquire continuous noble- and reactive-gas records (He, Ar, Kr, N_2, O_2, CO_2) over a one-year period. The instrument enabled unattended, on-site measurements with percent-level precision and high temporal resolution, well-suited to capture transient events. During the study, MAR operations ceased eight times for periods ranging from hours to several days. Each cease produced a potential characteristic multiphase gas response: (i) rapid shifts in dissolved gas ratios consistent with bubble entrapment and dissolution upon re-wetting, i.e., formation of "excess air"; (ii) step-changes in O_2 and N_2 that propagated on operational time scales; and (iii) knock-on effects on redox-sensitive species (CO_2, Ch_4) indicative of short-term biogeochemical reorganization. By combining noble-gas diagnostics of excess air with reactive-gas trends, we separate physical recharge effects from biogeochemical responses and quantify how shutdowns alter effective recharge signatures and residence-time proxies. These findings highlight portable mass spectrometry as a process-control tool for MAR, linking operational decisions to subsurface gas dynamics and water-quality outcomes in near-real time. Our results generalize established concepts of excess-air formation in porous media to the operational rhythms of MAR and provide actionable guidance for minimizing adverse water-quality swings during start/stop cycles.

XY Nov 2025, HH:MM–HH:MM

Dissolved Gas Monitoring at the Bedretto Underground Laboratory: Responses to Seismic Stimulation

Alexandra Lightfoot, Matthias Brennwald, Valentin Gischig, Alba Zappone, Rolf Kipfer, Stefan Wiemer

SED ETH Zürich

Gas Monitoring at the Bedretto Underground Laboratory: Responses to Seismic Stimulation In hydraulic stimulation experiments, such as those used in seismic and geothermal energy studies, monitoring dissolved reactive and noble gases in groundwater can provide key insights into fluid transport and redistribution. While tracers added to injection water reveal specific flow paths, dissolved gas monitoring in general captures a broader picture of how fluids are mobilised or redirected under pressure. At the Bedretto Underground Laboratory for Geosciences and Geoenergy (BULGG), we tested the geochemical response during an experiment designed to trigger a controlled magnitude-zero earthquake. High-pressure water (up to 20 MPa) was injected into borehole ST1, with krypton added as a tracer. A nearby borehole (ST2), 44 m away and discharging continuously, was equipped with a miniRuedi to measure dissolved He, Ar, Kr, N2, O2, and CO2 in real time, complemented by parallel radon monitoring. This setup enabled us to track both immediate gas responses to pressure and delayed tracer migration. Dissolved gas signals at ST2 revealed a clear response once injection began. N2 and Ar increased in terms of partial pressure, while He decreased, showing that injection resulted in mobilised or redirected younger, less radiogenic fluids toward ST2 well before the Kr tracer arrived. Radon activity increased only after tracer breakthrough, consistent with fracture opening and fluid movement under stress. These findings highlight dissolved gas monitoring as a sensitive tool for probing the interplay between stress, fractures, and fluids, particularly in the context of hydraulic stimulation.

XY Nov 2025, HH:MM–HH:MM

Noble gases as artificial tracers for routine hydrogeological investigations

Morgan PEEL, Kai SOLANKI, Daniel HUNKELER, Philip BRUNNER, Rolf KIPFER

Eawag

Noble gases are increasingly adopted as artificial tracers for hydro(geo)logical studies, thanks to advances in portable, high-resolution dissolved gas measurement technologies [1-3]. Their inert, non-toxic, and invisible nature makes them ideal for sensitive environments, such as rivers and drinking water catchment areas, where maintaining water quality and appearance is essential. The application of gases in aqueous environments poses unique challenges compared to traditional tracers, as potential degassing needs to be accounted for, and ideally avoided, during tracer injection. We present a straightforward method for preparing and injecting tracer solutions with accurately controlled dissolved gas concentrations, whilst avoiding tracer loss. A two-step degassing-pure gas saturation approach enables rapid preparation of high-concentration tracer solutions using readily available materials. We applied this method in a riverbank filtration setting to perform well-to-well tracer tests using helium-4 (He-4) and krypton-84 (Kr-84). Known quantities of both tracers were injected into observation wells upstream of a pumping well. A portable mass spectrometer ("miniRUEDI" GE-MIMS [3]) was used for continuous monitoring in the pumping well. Breakthrough curves of He-4 and Kr-84 enabled precise determination of groundwater flow velocities, travel times, and tracer recovery. These findings illustrate how noble gases complement traditional tracer methods in hydrogeological investigations. The method is adaptable to other gases (e.g., Ne, Kr, Xe, light hydrocarbons), significantly expanding the range of tracers available in groundwater management contexts. 1 : Brennwald et al., 2022, Front. Water (4) 2 : Blanc et al., 2024, Wat. Res. (254) 3 : Brennwald et al., 2016, Env. Sci. Techn. (50)

XY Nov 2025, HH:MM–HH:MM
TBD

I'm sure you could guess

University of Basel

I'll let you know as soon as I know, but it will be something about our continuous monitoring in Japan (+ some unsolicited but very funny jokes)

XY Nov 2025, HH:MM–HH:MM

TBD

AUTHORS

University College Dublin

Not yet sure - either something about in-situ depth profile measurements in a stratified pond, or preliminary IODP exp 501 results, or volcanic gas measurements in Costa Rica or Eifel, Germany

XY Nov 2025, HH:MM–HH:MM

Investigating gasgeochemical production during the preparatory phase of dynamic brittle failure in Rotondo granite in the laboratory

Paul Selvadurai, Claudio Madonna, Clement Roques, Hanna Ventura, Matthias Brennwald, Rolf Kipfler, Benoit, Valley, Alexandra Lightfoot, Laurent Marguet

ETH Zurich

Laboratory geochemistry studies enhance our understanding of earthquake precursors. Controlled experimental conditions allow researchers to observe how microcracks, fluid flow, and mineral reactions influence geochemical signals, including gas emissions and isotope distributions. These experiments can identify specific geochemical changes, such as the release of radon or noble gases, associated with rock fracturing. Such insights are crucial for developing indicators of large events while calibrating geochemical signals, establishing links between these signals, stress states, and damage levels within rocks.

This study investigates the use of helium, argon, and CO2 as real-time geochemical tracers for brittle deformation in crystalline rocks, focusing on Rotondo granite from the Bedretto Laboratory. These gases, stored in mineral lattices or fluid inclusions, are released during microstructural damage; however, their behavior during progressive rock failure remains poorly understood. Uniaxial compression experiments monitored gas variability with a miniRuedi quadrupole mass spectrometer, complemented by spatial strain measurements using Distributed Fibre Optic Strain (DFOS) sensing, acoustic emission (AE) monitoring, and post-mortem X-ray computed tomography (XRCT) of fracture networks. The experimental design aims to identify distinct deformation stages, from crack initiation to macroscopic failure, and link these to transient noble gas signals. Preliminary results reveal stage-specific fluctuations in helium and CO2, supporting the hypothesis that gas emissions reflect structural damage evolution. This research advances the integration of mechanical and geochemical monitoring in rock mechanics, with important implications for underground stability assessment and early warning systems in geotechnical and seismic environments.

XY Nov 2025, HH:MM–HH:MM

A (very) portable Gas Equilibrium Membrane Sampling System (GEMSS)

Yama Tomonaga, Oliver S. Schilling

Eawag / University of Basel

The miniRUEDI has proved to be a valuable tool to monitor continuously and in real time the gas dynamics in environmental systems. While mid- and long-term observations imply generally fixed installations with a relatively constrained effort in terms of logistics, the spatial screening of the gas composition in surface waters and groundwater requires the transport of miniRUEDI with the respective GE-MIMS (and a power source, e.g. a generator) to the different sampling locations. This can be sometimes problematic, as the instrument weight of about 30 kg (not including pumps, generator ...) makes it ”portable” only to a certain extent. To address this limitation, and to simultaneously improve on the long standing limitation of samples taken in copper tubes not being prescreenable on the miniRUEDI prior to time-consuming noble gas isotope analyses, we developed a small portable Gas Equilibrium Sampling System (GEMSS aka ”James”) that can be taken to the field easily and allows for a sufficient amount of gas to be sampled for multiple analyses on a single sample. The system uses the same concept of gas equilibrium of the conventional GE-MIMS. However, instead of functioning as an inlet for miniRUEDI measurements, James allows acquisition of gas samples for subsequent analyses in reusable containers. James is relatively light and fits in a backpack, it not only allows replicate measurements on a miniRUEDI, but the respective samples can further be used also for other types of analyses (e.g., conventional noble gas isotope analyses on magnetic sector mass spectrometers).

XY Nov 2025, HH:MM–HH:MM

On the hunt for freshwater under the oceanfloor (preliminary)

Paul Moser Röggla, Alizè Longeau, Israela Musan, Alan Seltzer, David Bekaert, Rolf Kipfer (preliminary)

Eawag

Freshened Offshore Groundwater off the Coast of New England Mini Ruedi for dissolved gases in submarine aquifer (Preliminary)

XY Nov 2025, HH:MM–HH:MM

ABSTRACT TITLE

BEN LEFEUVRE ET AL

AFFILIATIONS

ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT.

XY Nov 2025, HH:MM–HH:MM

ABSTRACT TITLE

PRESENTING AUTORS

AFFILIATIONS

ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT.

XY Nov 2025, HH:MM–HH:MM

ABSTRACT TITLE

JAKOB HELBING ET AL

AFFILIATIONS

ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT.

XY Nov 2025, HH:MM–HH:MM

Multi-Tracer Insights into Urban Catchment Dynamics and Wetland Support

Welham, A., van Rooyen, J., Watson, A., Chow, R., Roychoudhury, A.

Eawag

This contribution presents a multi-tracer study of an urbanized catchment in South Africa, combining isotopes (δ^{18}O, δ^{2}H), tritium, major ions, and dissolved natural gases to resolve interactions between rainwater, rivers, groundwater, and surface water transfer schemes. The results highlight how event-driven recharge, longer-residence groundwater, and managed surface flows jointly shape estuarine wetland resilience under climatic and anthropogenic pressures. (Will send the full abstract later today)

XY Nov 2025, HH:MM–HH:MM

New tool for in-situ gas measurements in low flow porous media

C. Marion, C. Ebi, S. Bloem, M.S. Brennwald, R. Kipfer

Eawag

Dummy dummy to be improved and checked by co-authors: Gases, including noble gases, are powerful tracers in environmental science, widely used to study groundwater flow dynamics and surface–subsurface water interactions (Brennwald et al. 2022, Blanc et al. 2024, Schilling 2017, Schilling 2021, Peel 2022 & 2023). However, most applications focus on high-flow systems (L/min to L/s), while in-situ gas measurements in low-flow porous media (L/h), such as soil unsaturated and saturated zones or plant tissues, remain technically challenging—despite their critical role in the global hydrological cycle. We present a robust, field-deployable gas probe for long-term, in-situ gas monitoring in low-permeability or low-flow environments. The device consists of a gas-tight, 3D-printed body with an integrated pressure sensor and gas-permeable membranes that allow diffusion of target gases into the probe's headspace for subsequent sampling and analysis. Originally developed for in-vivo gas monitoring in tree xylem (Marion et al., 2024), this method enables non-destructive, high-resolution measurements applicable to a variety of porous media, including soil, sediment, rock, and biological tissue.

XY Nov 2025, HH:MM–HH:MM

ABSTRACT TITLE

PRESENTING AUTORS

AFFILIATIONS

ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT.